



Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Phenomenology of the Standard Model of Elementary Particles

Master's Degree in Physics

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DI MILANO



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Probing the EW Sector and BSM Physics with $W^\pm Zjj$

1 Introduction

Experimental Setup & Target

- **Dataset:** 140 fb^{-1} of pp collisions at $\sqrt{s} = 13 \text{ TeV}$ (2015–2018).
- **Target:** Fully leptonic decays of $W^\pm Zjj$ production.

Analysis Objectives

- **Measurements:** Integrated & differential fiducial cross-sections (σ_{EW} and σ_{strong}).
- **Goal:** Isolate Vector Boson Scattering (VBS).

New Physics Search

- **BSM:** EFT limits on Anomalous Quartic Gauge Couplings (aQGC).



The ATLAS Detector

1 Introduction

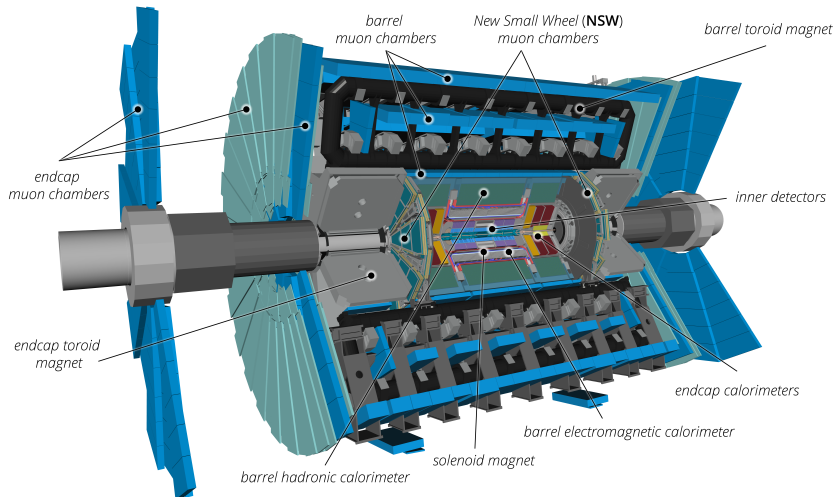




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Gauge Boson Self-Interactions

2 Theoretical Framework

- The Electroweak (EW) sector of the Standard Model is based on the **non-abelian** gauge symmetry $SU(2)_L \times U(1)_Y$.
- Unlike QED, the generators of $SU(2)_L$ do not commute. As a consequence, the physical vector bosons ($W^\pm, Z, \gamma$) **self-interact**.
- This structure directly dictates the existence of **Triple** (TGC) and **Quartic** (QGC) Gauge Couplings:

$$\mathcal{L}_{\text{TGC}} = \frac{1}{2} g \epsilon_{abc} W_b^\mu W_c^\nu (\partial_\mu W_{a\nu} - \partial_\nu W_{a\mu}) \quad \mathcal{L}_{\text{QGC}} = \frac{1}{4} g^2 \epsilon_{abc} \epsilon_{ars} W_b^\mu W_c^\nu W_{r\mu} W_{s\nu}$$

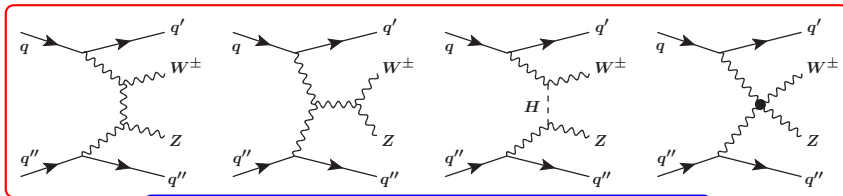
Probe these vertices to test the SM rigidity and search for anomalous couplings (aQGC) arising from **New Physics**.



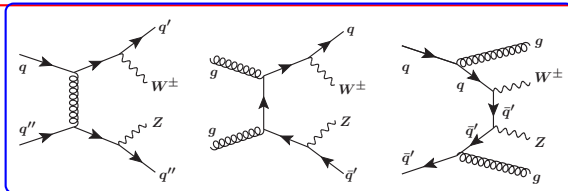
Experimental Signature: Vector Boson Scattering

2 Theoretical Framework

Vector Boson Scattering (VBS) $VV \rightarrow VV$ with $V = W/Z/\gamma$, is a key process with which to probe the $SU(2)_L \times U(1)_Y$ gauge symmetry of the **electroweak** (EW) theory that determines the **self-couplings** of vector bosons. To study these couplings, we exploit the purely electroweak production of $W^\pm Zjj$.



$W^\pm Zjj$ -EW



$W^\pm Zjj$ -QCD



A little look at the Effective Field Theory (EFT)

2 Theoretical Framework

The $W^\pm Zjj$ -EW production can be sensitive to effects beyond the SM affecting the quartic interactions of weak bosons. The measured $W^\pm Zjj$ events are therefore used to search for anomalous quartic gauge couplings (aQGC) using an **EFT framework**. In this model, the Lagrangian of the standard model is extended into an efficient Lagrangian by adding higher-order operators and their respective **Wilson coefficients** as follows:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i^{(6)}}{\Lambda^2} O_i^{(6)} + \sum_j \frac{f_j^{(8)}}{\Lambda^4} O_j^{(8)} + \dots$$

Where:

- $O_{i,j}^{(6),(8)}$: are dimension-6 and dimension-8 operators;
- $f_i^{(6)}$ e $f_j^{(8)}$: dimensionless couplings
- Λ : is the energy scale of the new processes



A little look at the Effective Field Theory (EFT)

2 Theoretical Framework

The squared scattering amplitude of the effective field theory prediction for $W^\pm Z jj$ production can be written as:

$$\left| A_{\text{SM}} + \sum_i c_i A_i \right|^2 = |A_{\text{SM}}|^2 + \sum_i c_i 2 \text{Re}(A_{\text{SM}}^* A_i) + \sum_i c_i^2 |A_i|^2 + \sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$$

- $c_i = f_j^{(8)} / \Lambda^4$, A_{SM} is the SM scattering amplitude;
- $2 \text{Re}(A_{\text{SM}}^* A_i)$ is the amplitude of the interference term between the SM and the dimension-8 operators;
- $\sum_i c_i^2 |A_i|^2$ is the pure dimension-8 contribution, and $\sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$ is the amplitude of interferences between two dimension-8 operators, called cross terms.



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Fiducial Phase Space for Leptons

3 Fiducial Phase Space and MC Simulations

The $W^\pm Z jj$ cross-sections are measured in a **fiducial phase space**. Leptons produced in the decay of a hadron, a τ -lepton, or their descendants are not considered in the definition of the fiducial phase space and are treated as background.

Category	Kinematic Variable	Requirement
Z leptons	Transverse momentum ($p_T^{\ell_Z}$)	$> 15 \text{ GeV}$
W lepton	Transverse momentum ($p_T^{\ell_W}$)	$> 20 \text{ GeV}$
All leptons	Pseudorapidity ($ \eta_\ell $)	< 2.5
Z boson	Invariant mass window ($ m_{\ell^+\ell^-} - m_Z^{\text{PDG}} $)	$< 10 \text{ GeV}$
W boson	Transverse mass (m_T^W)	$> 30 \text{ GeV}$
Angular isolation	Distance between Z leptons ($\Delta R(\ell_{Z1}, \ell_{Z2})$)	> 0.2
Angular isolation	Distance between W and Z leptons ($\Delta R(\ell_W, \ell_Z)$)	> 0.3



Fiducial Phase Space for Jets

3 Fiducial Phase Space and MC Simulations

In addition, at least **two jets are required**, reconstructed using the anti- k_t algorithm with radius parameter $R = 0.4$. Hard processes with a b -quark in the initial or final states, such as tZj production, are not considered as part of the $W^\pm Zjj$ -EW events.

Category	Kinematic Variable	Requirement
All jets	Transverse momentum (p_T^j)	$> 40 \text{ GeV}$
All jets	Pseudorapidity ($ \eta_j $)	< 4.5
Angular isolation	Distance between jets and leptons ($\Delta R(j, \ell)$)	> 0.3
Tagging jets	Opposite hemispheres ($\eta_{j1} \cdot \eta_{j2}$)	< 0
Tagging jets	Invariant mass (m_{jj})	$> 500 \text{ GeV}$
Jet Veto		
b -quark jets	Absence of b -jets in initial/final state	$N_{b\text{-jet}} = 0$



How to eliminate leptons ambiguity?

3 Fiducial Phase Space and MC Simulations

- Leptons and final-state neutrinos that do not originate from hadron or τ -lepton decays, are matched to the $W^\pm$ and Z boson decay products using an algorithmic approach, called the **resonant shape algorithm**.
- This algorithm is based on the value of an estimator expressing the product of the nominal line-shapes of the W and Z resonances:

$$P = \left| \frac{1}{m_{(\ell^+, \ell^-)}^2 - (m_Z^{\text{PDG}})^2 + i \Gamma_Z^{\text{PDG}} m_Z^{\text{PDG}}} \right|^2 \times \left| \frac{1}{m_{(\ell', \nu_{\ell'})}^2 - (m_W^{\text{PDG}})^2 + i \Gamma_W^{\text{PDG}} m_W^{\text{PDG}}} \right|^2$$



Simulation and Background Overview

3 Fiducial Phase Space and MC Simulations

Signal & $W^\pm Zjj$ Components

Generated with MG5_aMC@NLO + Pythia8:

- **EW ($W^\pm Zjj$ – EW):** Signal, purely electroweak (LO, $\mathcal{O}(\alpha_{EW}^6)$).
- **QCD ($W^\pm Zjj$ – QCD):** VBS background (NLO in QCD).
- **Interference:** EW-QCD quantum int. (LO), $+6 \div 12\%$ contribution in SR.

Other Backgrounds

- **Prompt Leptons:** ZZ , WW , VVV , $t\bar{t}V$, tZj .
Estimated via MC simulations.
- **Fake Leptons:** Misidentified jets.
Estimated via data-driven techniques.

Detector Simulation

All MC samples processed through Geant4 ATLAS detector simulation and standard reconstruction software.



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Event Selection: Physics Objects Definition

4 Event selection & Background estimation

Data Quality & Trigger:

- Only data with stable beams and fully operational sub-detectors.
- Single-lepton triggers used (Combined efficiency $\approx 99\%$ for offline-selected $W^\pm Zjj$ events).

Physics Objects Reconstruction:

- **Prompt Leptons** e, μ, ν (E_T^{miss}): Reconstructed combining ID tracks with ECAL clusters or MS tracks. Strict isolation and impact parameter criteria are applied to reject non-prompt leptons.
- **Hadronic Jets**: Clustered using the *anti- k_t* algorithm ($R = 0.4$).
- **b-tagging**: Deep-learning algorithms identify displaced vertices from b-hadron decays (crucial for top-quark veto).



Physics Objects: Leptons

4 Event selection & Background estimation

Object	Selection Criteria
Muons	<i>Medium</i> Identification (Inner Detector + Muon Spectrometer) $p_T > 15 \text{ GeV}$, $ \eta < 2.5$ Impact parameters: $ d_0/\sigma_{d_0} < 3.0$, $ z_0 \sin \theta < 0.5 \text{ mm}$ Isolation efficiency: 90 – 99% (depending on p_T)
Electrons	<i>Medium Likelihood</i> Identification $p_T > 15 \text{ GeV}$, $ \eta < 1.37$ or $1.52 < \eta < 2.47$ (crack region veto) Impact parameters: $ d_0/\sigma_{d_0} < 5.0$, $ z_0 \sin \theta < 0.5 \text{ mm}$ Isolation efficiency: 89 – 99% (depending on p_T)

*Note: Impact parameter cuts are crucial to select **prompt** leptons and reject non-prompt background from heavy-flavor decays.*



Physics Objects: Jets & E_T^{miss}

4 Event selection & Background estimation

Object		Selection Criteria
Jets		Clustered via <i>anti-k_t</i> algorithm ($R = 0.4$) $p_T > 25 \text{ GeV}$, $ \eta < 4.5$ Pile-up suppression (JVT): <i>tight</i> ($p_T < 60 \text{ GeV}$, $ \eta < 2.4$) and <i>loose</i> ($p_T < 120 \text{ GeV}$, $2.5 < \eta < 4.5$)
Missing Energy (E_T^{miss})		Proxy for the undetected neutrino. Computed as the negative vector sum of all calibrated <i>hard</i> objects (leptons, jets) plus a track-based <i>soft</i> term.



Event Selection: Boson Reconstruction

4 Event selection & Background estimation

Category	Selection Criteria
Baseline	Exactly 3 lepton candidates (μ or e) Veto on a 4th <i>loose</i> lepton ($p_T > 5$ GeV) Trigger matching: at least 1 lepton with $p_T > 25/27$ GeV
Z Boson Candidate	Same-Flavor Opposite-Sign lepton pair $ m_{\ell\ell} - m_Z^{\text{PDG}} < 10$ GeV (If ambiguous, the pair closest to m_Z is chosen)
W Boson Candidate	The remaining 3rd lepton + E_T^{miss} Stricter requirements: <i>Tight</i> ID & Isolation, $p_T > 20$ GeV Transverse mass: $m_T^W > 30$ GeV



Irreducible Backgrounds

4 Event selection & Background estimation

Irreducible backgrounds feature true prompt leptons mimicking the signal topology.

ZZ Control Region (ZZ-CR)

- **Background:** $ZZ \rightarrow 4\ell$ (ZZjj-QCD).
- **Origin:** Survives via lepton acceptance loss.
- **Selection:** Requires a 4th *loose* lepton to ensure orthogonality.

b Control Region (b-CR)

- **Background:** Top-quark ($t\bar{t} + V, tZj$).
- **Origin:** Survives via b-tagging inefficiency.
- **Selection:** Inverts the b-veto by requiring ≥ 1 b-jet.

While rare processes like **Tribosons** (VVV) and **ZZjj-EW** are estimated via **pure Monte Carlo** due to their low yields, the dominant sources are constrained using dedicated **Control Regions**:



Reducible Backgrounds (Fake Leptons)

4 Event selection & Background estimation

Unlike irreducible processes, reducible backgrounds in the $W^{\pm}Zjj$ phase space arise from instrumental misidentification rather than true prompt leptons.

Fake E_T^{miss} & Misidentified Objects

- **$Z + \text{jets}$:** A hadronic jet (e.g., charged pion) is misidentified as a lepton. E_T^{miss} is faked by finite Jet Energy Resolution.
- **$Z\gamma$:** A true photon mimics an electron.

True E_T^{miss} & Non-Prompt Leptons

- **$t\bar{t}$ & Wt :** True E_T^{miss} from $W \rightarrow \ell\nu$. The 3rd lepton is a non-prompt decay from a b -hadron inside a jet.
- **WW :** Two prompt leptons and true neutrinos. The 3rd lepton is faked by associated QCD jets or pile-up tracks.

Estimation Strategy: Data-Driven Matrix Method



Expected and Observed Numbers of Events Pre-Fit

4 Event selection & Background estimation

	SR ($N_{\text{jet}} = 2$)	SR ($N_{\text{jet}} \geq 3$)	$b\text{-CR}$	$ZZ\text{-CR}$
Data	169	477	666	210
Total pred.	231 ± 12	550 ± 50	660 ± 40	205 ± 11
$W^{\pm}Zjj\text{--EW}$	65.0 ± 3.5	60 ± 6	4.82 ± 0.28	0.725 ± 0.014
$W^{\pm}Zjj\text{--QCD}$	125 ± 9	380 ± 50	77 ± 18	6.2 ± 0.7
$W^{\pm}Zjj\text{-INT}$	1.3 ± 0.6	5.3 ± 2.6	0.58 ± 0.29	0.22 ± 0.11
$t\bar{t} + V$	0.66 ± 0.04	20.2 ± 0.7	289 ± 10	9.89 ± 0.28
tZj	8.78 ± 0.34	19.7 ± 1.2	134 ± 4	0.432 ± 0.005
$ZZ\text{-QCD}$	9.6 ± 0.4	32.0 ± 2.5	10.1 ± 0.6	159 ± 9
$ZZ\text{-EW}$	2.2 ± 0.6	4.4 ± 1.1	0.25 ± 0.06	23 ± 6
VVV	0.41 ± 0.10	2.0 ± 0.5	0.39 ± 0.10	4.1 ± 1.1
Misid. leptons	18 ± 4	28 ± 7	150 ± 40	1.7 ± 0.5



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Signal Extraction: Multivariate Analysis (BDT)

5 Measurement methodology and results

The Challenge

$WZjj$ -EW (signal, $\mathcal{O}(\alpha_{EW}^6)$) and $WZjj$ -QCD (background, $\mathcal{O}(\alpha_s^2 \alpha_{EW}^4)$) share the exact same final state. Kinematic cut-based separation is sub-optimal.

⇒ **Solution:** Boosted Decision Tree (TMVA).

BDT Input Variables (15 observables):

- *Jet*: m_{jj} , $|\Delta\eta_{jj}|$, $\Delta\phi_{jj}$, N_{jets}
- *Boson*: p_T^W , p_T^Z , m_T^{WZ}
- *Correlations*: Lepton centrality, event balance

BDT Output:

- Maps a multi-dimensional phase space into a **single discriminant score**.
- Used as the primary observable for signal extraction.

Note: An independent auxiliary BDT is deployed in the b -CR to differentiate between $t\bar{t}V$ and tZj .



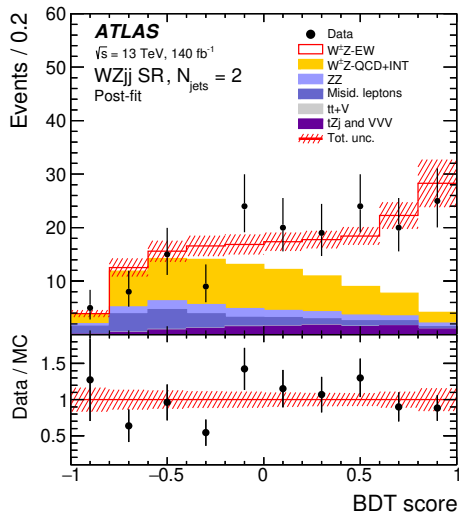
The Global Likelihood Fit & Post-fit distributions SR

$$(N_{jets} = 2)$$

5 Measurement methodology and results

Fit Components

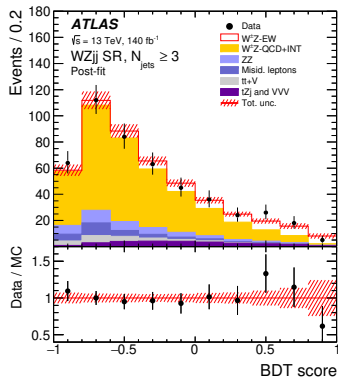
- **Regions (r):** SR ($N_{jets} = 2$), SR ($N_{jets} \geq 3$), b -CR, ZZ-CR.
- **Parameters of Interest (μ):** Strength modifiers μ_{EW} and μ_{QCD} .
- **Unconstrained Parameters (K):** Background normalizations ($K_{t\bar{t}V}$, K_{tZj} , K_{ZZ}) extracted dynamically from CRs.
- **Nuisance Parameters (ν):** Theoretical/experimental systematics, constrained by a Gaussian penalty term.



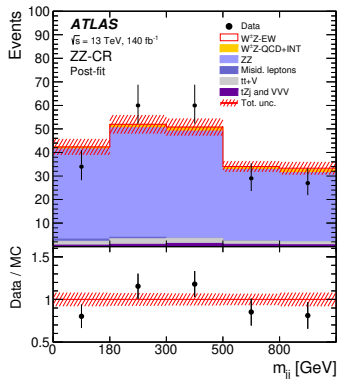


Post-fit distributions SR ($N_{jets} \geq 3$), ZZ-CR and b-CR

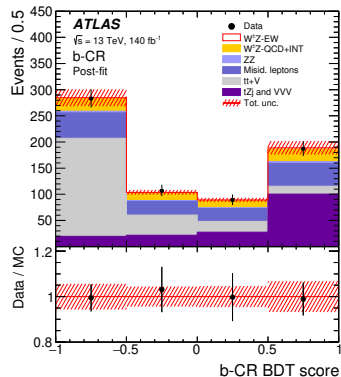
5 Measurement methodology and results



(a)



(b)



(c)



Fiducial Cross-Section Extraction

5 Measurement methodology and results

The fiducial cross-sections are extracted directly from the profile-likelihood fit. The normalisation factors (μ_{EW} , μ_{QCD}) scale the theoretical MC predictions within the fiducial phase space:

$$\sigma_{WZjj-EW} = \sum_i \mu_i^{WZjj-EW} \cdot \sigma_{i,th.MC}^{WZjj-EW}$$

$$\sigma_{WZjj-strong} = \sum_i \left(\mu_i^{WZjj-QCD} \cdot \sigma_{i,th.MC}^{WZjj-QCD} + \mu_i^{WZjj-INT} \cdot \sigma_{i,th.MC}^{WZjj-INT} \right)$$

*Index i runs over the two SR categories: $N_{jets} = 2$ and $N_{jets} \geq 3$.

- **Interference term:** μ_{INT} is constrained as $\sqrt{\mu_{EW}} \cdot \sqrt{\mu_{QCD}}$.



Results and Comparison with SM Predictions

5 Measurement methodology and results

VBS (EW Production)

$$\sigma_{\text{obs}}^{\text{EW}} = 0.368 \pm 0.037_{\text{stat}} \pm 0.059_{\text{syst}} \text{ fb}$$

$$\sigma_{\text{th}}^{\text{EW}} = 0.370 \pm 0.030 \text{ fb}$$

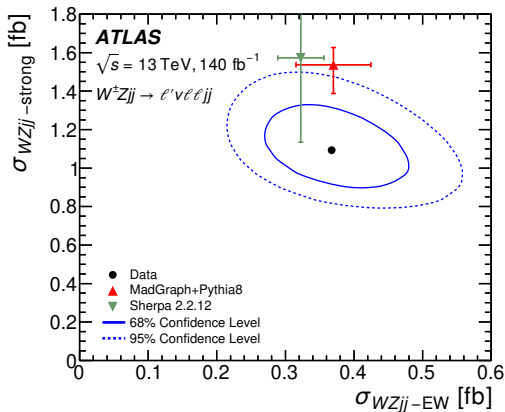
⇒ Excellent SM agreement

Strong + Int.

$$\sigma_{\text{obs}}^{\text{strong}} = 1.093 \pm 0.066_{\text{stat}} \pm 0.131_{\text{syst}} \text{ fb}$$

$$\sigma_{\text{th}}^{\text{strong}} = 1.537 \pm 0.150 \text{ fb}$$

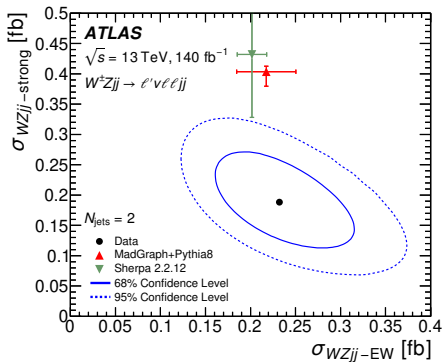
⇒ Deficit factor ~ 0.7 (1.8σ tension)



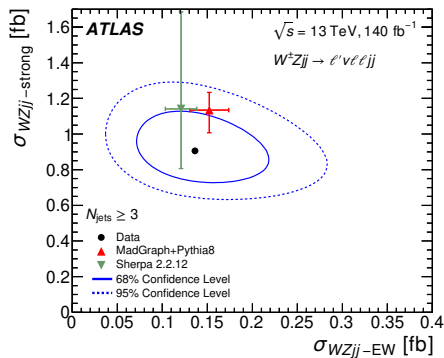


$W^\pm Zjj$ – EW and $W^\pm Zjj$ -strong ($N_{jets} = 2$), SR ($N_{jets} \geq 3$)

5 Measurement methodology and results



(a)

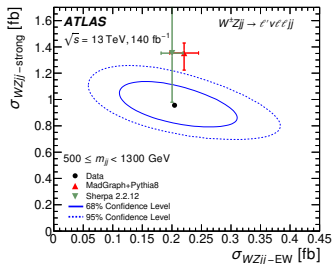


(b)

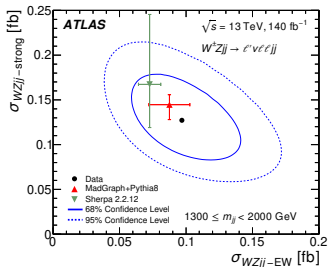


$W^{\pm}Zjj$ – EW and $W^{\pm}Zjj$ -strong in different m_{jj} bins

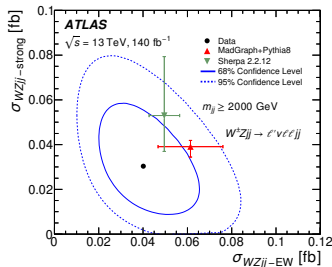
5 Measurement methodology and results



(a)



(b)



(c)



$W^\pm Z jj$ differential measurements

5 Measurement methodology and results

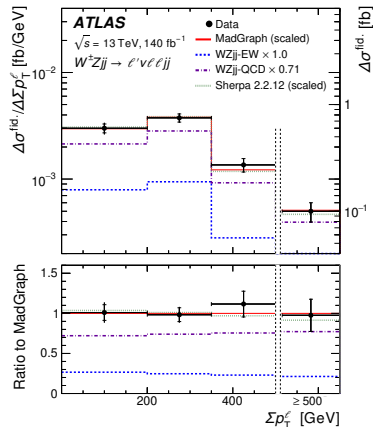
Differential measurements of $W^\pm Z jj$ production cross-section are performed as a function of variables sensitive to anomalies in the quartic gauge coupling.

Kinematic Variables

$\sum \mathbf{p}_T^\ell$ Scalar sum of the transverse momenta of the three charged leptons (W , Z).

$\Delta\phi(\mathbf{W}, \mathbf{Z})$ Difference in azimuthal angle between the W and Z bosons' directions.

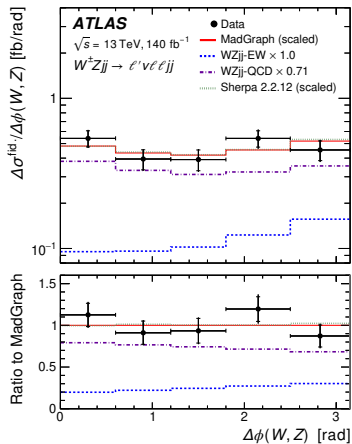
m_T^{WZ} Transverse mass of the $W^\pm Z$ system.



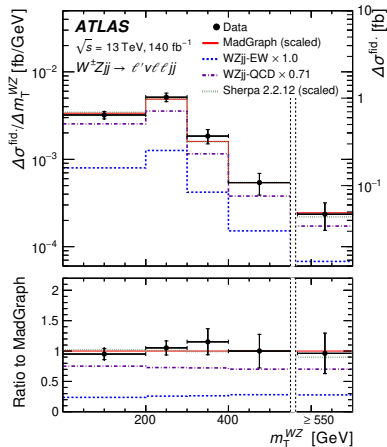


$W^{\pm}Zjj$ differential measurements

5 Measurement methodology and results



(a)



(b)



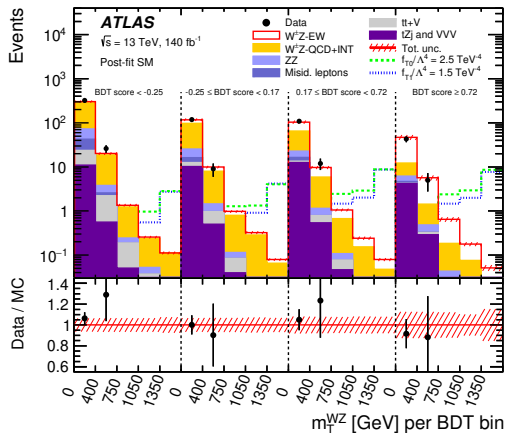
Search for aQGC: Methodology

5 Measurement methodology and results

Anomalous Quartic Gauge Couplings modeled via **EFT dimension-8 operators** drastically enhance the cross-section at high energy scales ($\hat{s}$).

2D Fit Strategy

- **Observables:**
BDT score (EW vs QCD) & m_T^{WZ} .
- **Binning:**
 $4 \text{ BDT} \times 5 m_T^{WZ} \rightarrow$ **1D histogram** (20 bins).
- **Test Statistic:**
Profiled Likelihood Ratio $q(\vec{c})$ for **95% CL limits** on $f_j^{(8)} / \Lambda^4$.





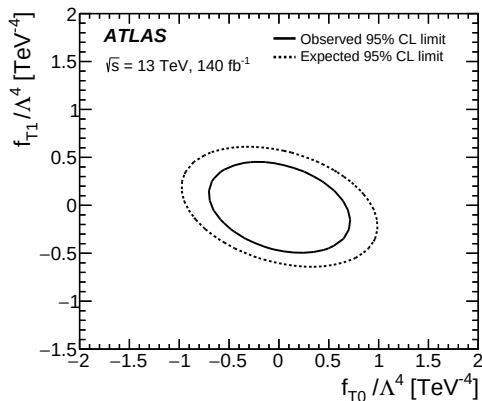
EFT Limits (Without Unitarization)

5 Measurement methodology and results

1D Limits (95% CL)

Coupling	Exp. [TeV^{-4}]	Obs. [TeV^{-4}]
f_{T0}/Λ^4	$[-0.80, 0.80]$	$[-0.57, 0.56]$
f_{T1}/Λ^4	$[-0.52, 0.49]$	$[-0.39, 0.35]$
f_{T2}/Λ^4	$[-1.6, 1.4]$	$[-1.2, 1.0]$
f_{M0}/Λ^4	$[-8.3, 8.3]$	$[-5.8, 5.6]$
f_{M1}/Λ^4	$[-12.3, 12.2]$	$[-8.6, 8.5]$
f_{M7}/Λ^4	$[-16.2, 16.2]$	$[-11.3, 11.3]$
f_{S02}/Λ^4	$[-14.2, 14.2]$	$[-10.4, 10.4]$
f_{S1}/Λ^4	$[-42, 41]$	$[-30, 30]$

2D Contour (f_{T0} vs f_{T1})





Tree-Level Unitarity Violation and Clipping

5 Measurement methodology and results

The EFT is not a complete model. Non-zero dimension-8 operators violate tree-level unitarity at high energies.

The "Clipping" Technique

- EFT contributions are removed above a cut-off scale E_c ($m_{WZ} > E_c$).
- Replaced by SM predictions to restore unitarity.

Unitarized Limits (at crossing bound):

- $f_{T0}/\Lambda^4 \in [-1.5, 1.6] \text{ TeV}^{-4}$
- $f_{T1}/\Lambda^4 \in [-0.7, 0.6] \text{ TeV}^{-4}$

Evolution of 95% CL limits vs cut-off scale $\Rightarrow$

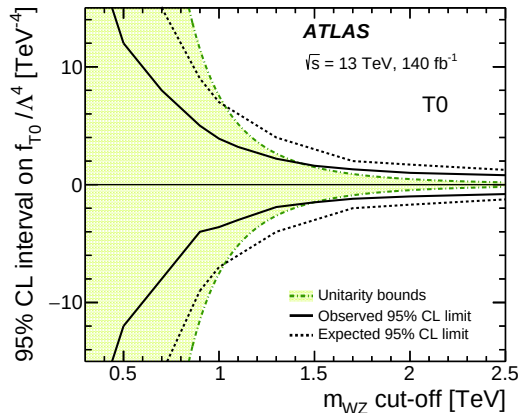




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Conclusions: VBS as a Precision Probe for the EWK Sector

6 Conclusion

From Discovery to Precision Measurement (2015-2018, 140 fb^{-1})

- **Integrated cross-section:** $\sigma_{WZjj\text{-EW}} = 0.368 \pm 0.037 \text{ (stat)} \pm 0.059 \text{ (syst) fb.}$
- **Differential measurements** (via Bayesian Unfolding) on aQGC-sensitive variables.

Standard Model Verification and Modelling Challenges

- **EW Production:** Excellent Data/Theory agreement in the Signal Region.
- **QCD Background:** Tension of 1.8σ (Data $\sim 30\%$ lower than nominal MadGraph).

Limits on New Physics (EFT) and Unitarity

- **95% CL limits** set on 9 (8) dimension-8 operators without (with) consideration of tree-level unitarity violation.
- These constraints are similar to those obtained by the CMS Collaboration.



Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Thank you for listening!
Any questions?



BACKUP



Monte Carlo Simulation of the Signal

6 Conclusion

The simulation of $W^{\pm}Zjj$ events was primarily generated using the MadGraph5_aMC@NL02.6.5 MC event generator, interfaced with Pythia8.240. The simulation was divided into distinct components:

- **Electroweak production $W^{\pm}Zjj$ – EW:** The actual signal. It is calculated at leading order (LO), considering purely electroweak processes (order α_{EW}^6).
- **QCD production $W^{\pm}Zjj$ – QCD:** Considered a background for the VBS analysis. This component is calculated at next-to-leading order (NLO) in QCD.
- **Interference $W^{\pm}Zjj$ – INT:** Quantifies the quantum interference between the EW and QCD production. It is calculated at LO and yields a positive contribution of 6% in the fiducial region, reaching up to 12% for events with three or more jets.



Background and Detector Simulation

6 Conclusion

In addition to the $W^{\pm}Zjj$ – QCD production, there are other processes that can mimic the signal in the detector.

- **Backgrounds with *prompt* (real) leptons:** Processes such as the production of ZZ pairs, WW , tribosons (VVV), or top quarks associated with vector bosons ($t\bar{t}V$, tZj). Their contribution is evaluated entirely using Monte Carlo simulations.
- **Backgrounds with *fake* leptons:** Events where a hadronic jet is misidentified as a lepton by the detector. Estimated directly using data-driven techniques.

All generated MC events were passed through the ATLAS detector simulation, based on Geant4, and processed using the same reconstruction software used for the data.



The Global Likelihood Fit

6 Conclusion

Simultaneous Extended Binned Negative Log-Likelihood:

$$-\ln \mathcal{L}(\mu, \mathbf{K}, \nu) = \sum_r \sum_i \left(\mu_{\text{exp}}^{[r,i]} - N^{[r,i]} \ln \mu_{\text{exp}}^{[r,i]} \right) + \sum_j \frac{\nu_j^2}{2}$$

- **Regions (r):** SR ($N_{\text{jets}} = 2$), SR ($N_{\text{jets}} \geq 3$), b -CR, ZZ-CR.
- **Observables ($N^{[r,i]}$):** bins BDT score in SRs and in b -CR and m_{jj} in ZZ-CR.
- **Parameters of Interest (μ):** Strength modifiers μ_{EW} and μ_{QCD} .
- **Unconstrained Parameters ($\mathbf{K}$):** Background normalizations ($K_{t\bar{t}V}$, K_{tZj} , K_{ZZ}) extracted dynamically from CRs.
- **Nuisance Parameters (ν):** Theoretical/experimental systematics, constrained by the Gaussian penalty term $\nu_j^2/2$.